KSME

Springer

Journal of Mechanical Science and Technology 39 (8) 2025

Original Article
DOI 10.1007/s12206-025-0445-5

Keywords:
· Heavy-load
· Manipulator
· Dynamics modeling
· Semi-parametric
· Deep learning

Correspondence to:
Shoujun Wang
wangshoujun@tjut.edu.cn

Citation:
Liu, N., Li, X., Wang, S. (2025). Inverse dynamics modeling of 7-DOF heavy-duty hydraulic manipulator. Journal of Mechanical Science and Technology 39 (8) (2025) 4671~4684.
http://doi.org/10.1007/s12206-025-0445-5

Received April 11th, 2024
Revised December 23rd, 2024
Accepted February 5th, 2025

† Recommended by Editor Dongho Oh

© The Korean Society of Mechanical Engineers and Springer-Verlag GmbH Germany, part of Springer Nature 2025

Inverse dynamics modeling of 7-DOF heavy-duty hydraulic manipulator

Nan Liu[1,2], Xingguo Li[1,2] and Shoujun Wang[1,2]

[1]National Demonstration Center for Experimental Mechanical and Electrical Engineering Education, Tianjin University of Technology, Tianjin 300384, China, [2]Tianjin Key Laboratory for Advanced Mechatronic System Design and Intelligent Control, School of Mechanical Engineering, Tianjin University of Technology, Tianjin 300384, China

Abstract A multi-feature semi-parametric modeling method is proposed aiming at the inverse dynamics modeling problem of the 7-DOF heavy-duty hydraulic manipulator (HDHM), which is widely applied in construction machinery. The method can compensate for the influence of nonlinear factors on manipulator inverse dynamics modeling, and further improve the accuracy and generalization ability of the dynamics model. First, the construction process of the parametric model is de-scribed in detail. Then, based on recurrent neural network (RNN), a multi-feature semi-parametric dynamics model (MSDM) is constructed and validated under two operating conditions (high-speed and medium-load, high-speed and heavy-load) of the 7-DOF HDHM. The experimental results show that the MSDM model performs optimally compared to the other models mentioned in this paper, laying the foundation for path optimization and control of the 7-DOF HDHM.

1. Introduction

Heavy-duty hydraulic manipulators are characterized by high power density, compact structure, good stiffness, fast response, and so on. They have a more compact mechanical structure, stronger loading capacity, and larger workspace compared to electrically driven manipulators [1]. Therefore, multi-degree-of-freedom heavy-duty hydraulic manipulators have been widely used in forestry, disaster rescue, mining, tunnelling and machining assembly [2-4]. In recent years, with the gradual increase in demand for heavy-duty hydraulic manipulators, higher requirements have been put forward for their load capacity and positioning accuracy [5, 6]. In order to achieve high-speed, heavy-load and high accuracy, the multi-degree-of-freedom hydraulic manipulator system is required to have good dynamics characteristics. And in order to obtain accurate dynamics characteristics, it is necessary to establish an accurate and dynamics model which can be quickly solved. However, the nonlinear and complex coupling time-varying characteristics arising from the joint action of multi-degree of freedom, high-speed, heavy-load, and hydraulic drive have put forward more challenges to its dynamics modeling [7].

Dynamics modeling is an important aspect of the study of heavy-duty hydraulic manipulators, providing the foundation for response prediction, motion control and parameter optimization of manipulators. High-speed, heavy-load operating conditions and high load to weight ratio cause large inertial forces of the manipulator when operating, leading to severe joint flexibility effects [8]. The compressibility and leakage of the hydraulic fluid also make the joint flexibility more significant and complex, increasing the difficulty of dynamics modeling. The accuracy of the manipulator dynamics system modeling can be affected by the nonlinear relationship between the input and output variables of the hydraulic system, the pressure and flow rate fluctuations caused by the differences in component characteristics, load variations, and piping layouts, and the hydraulic oil contamination, leakage, and fluid temperature variations [9, 10]. The modeling accuracy of the manipulator is affected by steady-state tracking errors, limit loop oscillations, low-speed creep, and waveform distortion at speeds past zero caused by nonlinear rubbing affect. And high-speed, heavy-load, and the sealing design of the hydraulic piston lead to a

more significant impact of nonlinear rubbing on the accuracy of the dynamics modeling [11-13].

The current problem of modeling the dynamics of manipulators can be divided into three main model types: parametric, non-parametric and semi-parametric models [14]. Whether a model is parametric or not depends on whether parameters based on known physics are used. Low et al. [15] investigated a method for deriving the dynamics equations of a manipulator containing both rigid and flexible linkages by extending the Lagrangian method. Pan et al. [16] used the Lagrangian dynamics modeling method to establish the dynamics model of a robot from the energy perspective of a six-degree-of-freedom robot and further verified the correctness of the dynamics model using simulation software. Bascetta et al. [17] established a closed dynamics model of a flexible manipulator based on the Newton-Euler formulation of the flexible linkage motion using a space vector representation. Zhang et al. [18] proposed a simplified dynamics modeling method, which greatly reduced the order and complexity of the closed-form dynamics equations. Yang et al. [19] proposed an efficient recursive solution method based on the lie group (RLGA), which can greatly improve the computational efficiency and be more easily applied when dealing with elastic nodes in inverse dynamics analysis compared with the traditional D-H method. Nonparametric methods are usually data-driven, such as Gaussian process regression, neural networks and Gaussian mixture models [20, 21]. The data-driven nature of the algorithms can lead to problems if the data is of poor quality or limited in quantity, which can also affect the generalizability of the model. Nguyen-Tuong and Peters et al. [22, 23] evaluated the suitability of three nonparametric methods (locally weighted projection regression (LWPR), support vector machine (SVM), and Gaussian process regression (GPR)) for learning dynamics, followed by an overview of research advances in robotic model learning in terms of both kinematics and dynamics. Semi-parametric method combines the characteristics of parametric and nonparametric methods to complete the modeling of robot dynamics. It is usually compensated by the rigid body dynamics formulation in the parametric method combined with certain functions in the nonparametric method, which can make up for the absence of some nonlinear factors in the modeling of the conventional parametric method. Romeres et al. [24] proposed a semi-parametric method for online learning of inverse dynamics models of robots by combining the advantages of parametric and non-parametric modeling, using rigid body dynamics equations and appropriate kernel functions. Camoriano et al. [25] proposed a new incremental semiparametric inverse dynamics modeling approach that combines parametric and nonparametric model learning in an incremental manner.

It can be seen that the traditional modeling approach requires accurate physical models and parameters, and the main advantage of this approach is that it provides a global relationship between inputs (joint angles, velocities and accelerations) and outputs (torques). However, linear models cannot capture nonlinearities in the data. The nonparametric approach overcomes the reliance on an exact model by allowing the model to learn any nonlinear function that encapsulates all and any forces present in the robot dynamics. The accuracy of the model may be low if the data is insufficient or of poor quality, which can also affect the generalizability of the model. The parametric part of the semi-parametric method is designed to deal only with conventional rigid body dynamics, while the nonparametric part is designed to represent nonlinear errors in all forces not defined in the parametric part. In order to establish an accurate dynamics model of 7-DOF heavy-duty hydraulic manipulator, it is necessary to take into account the effects of the flexible and nonlinear rubbing that exists between the actuator and the moving joints under high-speed and heavy-load operating conditions, as well as the effects of nonlinearities and complex coupling time-varying characteristics brought about by the multiple-degree-of-freedom and the hydraulic system on the basis of the rigid body dynamics of the manipulator system. Therefore, if a priori knowledge is used to design the extraction equations for the parametric features of joint rubbing (JR) and joint flexibility (JF), and JR and JF together with rigid body dynamics (RBD) as the parametric part, it is a feasible solution to establish the dynamics model of 7-DOF heavy-duty hydraulic manipulator in conjunction with the nonparametric method. However, we have not yet seen any research on inverse dynamics modeling by combining RBD, JR and JF together as parametric parts and deep learning-based nonparametric method.

Therefore, based on the above study, this paper designs the extraction equations for the parametric features of RBD, JR and JF by using a priori knowledge. Then, based on recurrent neural network (RNN), a multi-feature semi-parametric dynamics model (MSDM) is constructed of the 7-DOF heavy-duty hydraulic manipulator (HDHM). The structure of the paper is as follows: in Sec. 2, multi-feature semi-parametric modeling method based on deep learning is proposed and the extraction process of RBD, JR, and JF parametric features are presented. In Sec. 3, the multi-feature semi-parametric dynamics modeling framework of 7-DOF HDHM is introduced. Sec. 4 introduces the experimental platform and the method of obtaining data set. Sec. 5 develops the corresponding experiments and discusses the results. Finally, Sec. 6 makes the conclusion and proposes the direction of the future study.

2. Method

In this section, the multi-feature semi-parametric modeling approach, the establishment of extraction equations for the RBD, JR and JF parametric features, and the multi-feature semi-parametric dynamics model construction strategy for the 7-DOF HDHM will be presented in detail.

2.1 Construction method of multi-feature semi-parameter dynamics model

To address the problem that only RBD is used in the pa-

rameter part of the semi-parametric modeling method, while the influence of JR and JF on the accuracy of the dynamics model is ignored. Making full use of the a priori knowledge of rubbing and flexibility theory, we extracted and added JR and JF features to the parameter part, and the multi-feature semi-parametric modeling method was proposed. To make RBD, JR, JF parameterized features as the parametric dataset of multi-feature semi-parametric dynamics model (MSDM) and provide parameterized features for the non-parametric dataset, in order to reduce the demand of model on the amount of training data, and to improve the accuracy and generalization of the 7-DOF HDHM dynamics model. The specific form of the method is shown in Eq. (1).

$$\tau_{MSDM} \sim MSDM(q,\dot{q},\ddot{q},\tau_{RBD},\tau_{JR},\tau_{JF}) \sim SDM(q,\dot{q},\ddot{q},\tau_{RBD}) \sim NDM(q,\dot{q},\ddot{q}). \quad (1)$$

In Eq. (1), NDM is the non-parametric dynamics model, $q,\dot{q},\ddot{q}$ is the displacement, velocity and acceleration of each joint of the 7-DOF HDHM, which are the non-parametric data obtained through sensor acquisition, and are taken as inputs to the NDM. SDM is the semi-parametric dynamics model, τ_{RBD} is the RBD parametric data calculated through the rigid body dynamics equations, and $q,\dot{q},\ddot{q}$ and τ_{RBD} are taken as the inputs to the SDM together. $MSDM$ is the multi-featured semiparametric dynamics model. τ_{JR} and τ_{JF} are the JR and JF parameter data computed through the equations, respectively, and $q,\dot{q},\ddot{q}$, τ_{RBD}, τ_{JR}, and τ_{JF} are jointly taken as inputs to the $MSDM$. τ_{MSDM} is the prediction result of the $MSDM$, representing the force/moment at each joint of the 7-DOF HDHM.

As can be seen from Eq. (1), the parameter part of multi-feature semi-parametric modeling method makes full use of the RBD features and adds FR and JF features to compensate for the effects of joint rubbing and joint flexibility on the modeling of 7-DOF HDHM dynamics. This is then combined with a non-parametric component to fit non-linear features that have been neglected or simplified in the modeling process of parametric methods. The key properties of multi-feature semi-parametric modeling method are: making full use of the theoretically mature and rapidly solvable parts of the parametric model. An appropriate deep learning framework is chosen for the non-parametric part. It is possible to optimize both the parametric and non-parametric parts of the model at the same time, i.e., the weights of both the parametric and non-parametric parts will be continuously optimized during the training process.

From the above, the performance of a multi-feature semi-parametric deep learning model is simultaneously determined by both the parametric and non-parametric components :

1) When some state spaces (e.g. high-speed, heavy-load, etc.) exist, the nonlinear and complex coupling characteristics brought about by high-speed and heavy-load will lead to a decrease in the modeling accuracy of the parametric part, and then the nonparametric part will play a role in order to improve the overall performance of the multi-feature semi-parametric model.

2) In a state space where training data is relatively scarce, the non-parametric part will not be able to perform the deep learning network fitting ability well due to insufficient training, and the modeling accuracy will be mainly determined by the parametric part.

3) If both the parametric and non-parametric parts work well, it is then possible to improve the accuracy of the dynamics modeling, strengthen the generalization ability of the model, and reduce the amount of data required for a multi-feature semi-parametric model.

2.2 Constructing strategy of multi-feature semi-parametric dynamics model

The construction strategy of the complete 7-DOF HDHM multi-feature semi-parametric dynamics model is shown in Fig. 1, including training data acquisition, parametric feature extraction, training data processing, network architecture design,

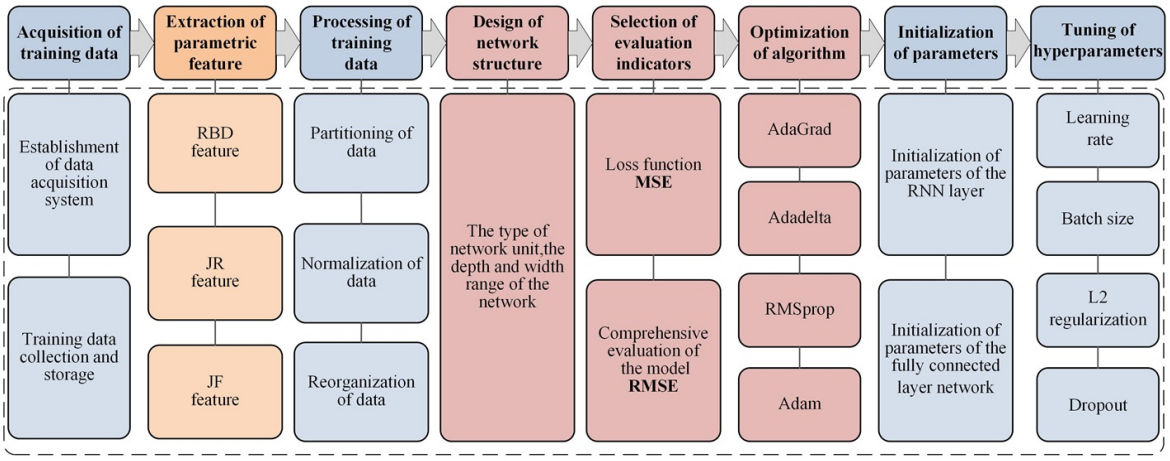


Fig. 1. Construction strategy and process of multi-feature semi-parametric dynamics model for 7-DOF HDHM.

selection of evaluation metrics, optimization algorithm, parameter initialization and hyperparameter tuning.

The features are numerical representations of the raw data, and the raw input data for the 7-DOF HDHM are the position, velocity, and acceleration of each joint, with $\boldsymbol{q}=[q_1, q_2, q_3, q_4, q_5, q_6, q_7]^{\mathrm{T}} \in \mathbb{R}^{7\times1}$ being the joint position, i.e., the generalized coordinate vector. $\dot{\boldsymbol{q}}=[\dot{q}_1, \dot{q}_2, \dot{q}_3, \dot{q}_4, \dot{q}_5, \dot{q}_6, \dot{q}_7]^{\mathrm{T}} \in \mathbb{R}^{7\times1}$ is the joint velocity, $\ddot{\boldsymbol{q}}=[\ddot{q}_1, \ddot{q}_2, \ddot{q}_3, \ddot{q}_4, \ddot{q}_5, \ddot{q}_6, \ddot{q}_7]^{\mathrm{T}} \in \mathbb{R}^{7\times1}$ is the joint acceleration. Feature engineering is the process of designing the most appropriate parametric features given the data, model and task, and the methodology for extracting parametric features using 7-DOF HDHM raw data is described in detail below.

2.2.1 Extraction of rigid body dynamics (RBD) parametric feature

The rigid body dynamics (RBD) features are the joint forces/torques of each joint of the 7-DOF HDHM obtained by using Newtonian Eulerian recursive derivation [26]:

$$D(q)\,\ddot{q}+B(q)[\dot{q}\,\dot{q}]+C(q)[\dot{q}^2]+f(\dot{q})=\tau\,. \tag{2}$$

In Eq. (2), $\boldsymbol{q}=[q_1, q_2, \ldots, q_7]^{\mathrm{T}} \in \mathbb{R}^{7\times1}$ is the position of each joint of the 7-DOF HDHM, because joints 1, 2, 4, 6, and 7 are rotary joints, while joints 3 and 5 are moving joints, so q_1, q_2, q_4, q_6, q_7 are the angle of rotation of the corresponding joints. q_3 and q_5 are the displacements of joints 3 and 5, respectively. $\dot{\boldsymbol{q}}=[\dot{q}_1, \dot{q}_2, \ldots, \dot{q}_7]^{\mathrm{T}} \in \mathbb{R}^{7\times1}$ is the velocity of each joint of the 7-DOF HDHM; $\ddot{\boldsymbol{q}}=[\ddot{q}_1, \ddot{q}_2, \ldots, \ddot{q}_7]^{\mathrm{T}} \in \mathbb{R}^{7\times1}$ is the acceleration of each joint of the 7-DOF HDHM. $\boldsymbol{\tau}=[\tau_1, \tau_2, \ldots, \tau_7]^{\mathrm{T}} \in \mathbb{R}^{7\times1}$ is the force/torque of each joint of the 7-DOF HDHM obtained by the Newtonian Eulerian recursive method based on the rigid body dynamics.

Extracting $\tau_1, \tau_3, \tau_4, \tau_5, \tau_6, \tau_7$ as a parametric feature, the joint torque τ_2 of joint 2 needs to be converted by vector mechanics to the driving force of the hydraulic cylinder of joint 2:

$$T_2=\frac{\tau_2\,{}^2H_2^{\mathrm{R}}}{{}^2\overrightarrow{OA'}\times{}^2\boldsymbol{e}} \tag{3}$$

$${}^2\boldsymbol{e}=\frac{{}^2\overrightarrow{BA'}}{|{}^2\overrightarrow{BA'}|}=\frac{{}^2\overrightarrow{BA'}}{\sqrt{{}^2\overrightarrow{BA'}\cdot{}^2\overrightarrow{BA'}}} \tag{4}$$

$${}^2\overrightarrow{BA'}={}^2\overrightarrow{OA'}-{}^2\overrightarrow{OB'}={}^2\overrightarrow{OA'}-{}^2_1\boldsymbol{A}\,{}^1\overrightarrow{OB'}\,. \tag{5}$$

Using the converted hydraulic cylinder drive T_2 as a parameterized feature.

In summary, the original features of the 7-DOF HDHM:

$$\boldsymbol{X}_{Ori}=[q_1, \dot{q}_1, \ddot{q}_1, q_2, \dot{q}_2, \ddot{q}_2, \cdots, q_7, \dot{q}_7, \ddot{q}_7]^{\mathrm{T}} \in \mathbb{R}^{21\times1}\,. \tag{6}$$

Based on Eq. (6), the RBD feature computation of the 7-DOF HDHM was carried out using Eqs. (2)-(5), and the obtained $\tau_1, T_2, \tau_3, \tau_4, \tau_5, \tau_6, \tau_7$ was used as the RBD parameterized feature of the MSDM model:

$$\boldsymbol{X}_{RBD}=[\tau_1, T_2, \tau_3, \tau_4, \tau_5, \tau_6, \tau_7]^{\mathrm{T}}\,. \tag{7}$$

In Eq. (7), $\boldsymbol{X}_{RBD} \in \mathbb{R}^{7\times1}$ is the RBD parameterized feature of the MSDM model.

2.2.2 Extraction of joint rubbing (JR) parametric feature

In order to more accurately establish the dynamics model of the 7-DOF HDHM, a static continuous rubbing model based on smooth tanh function proposed by Makkar et al. [27] is used to establish the rubbing model of each joint, which can eliminate the undesirable effects caused by the discontinuity of the rubbing model.

$$\begin{aligned} f(\dot{q}_i)&=\gamma_1[\tanh(\varphi_1\dot{q}_i)-\tanh(\varphi_2\dot{q}_i)]+\gamma_2\tanh(\varphi_3\dot{q}_i)\\ &+\gamma_3\dot{q}_i+\varphi \quad (i=1\sim7)\,. \end{aligned} \tag{8}$$

In Eq. (8), $\dot{q}_i$ is the velocity of the 7-DOF HDHM joint i; for joints 1, 4, 6 and 7, $\dot{q}_i$ is the joint angular velocity. For joints 2, 3 and 5, $\dot{q}_i$ is the joint linear velocity γ_1, γ_2 and γ_3 represent the different degrees of rubbing, and φ_1, φ_2 and φ_3 denote the various shape factors to approximate various rubbing effects.

In order to obtain the rubbing force of each joint of the 7-DOF HDHM by Eq. (8), the rubbing model parameters need to be identified by means of experiments :

1) A measurement system for the rubbing force of each joint of the 7-DOF HDHM was built and rubbing force measurement tests were conducted. Since the rubbing force cannot be measured directly but only indirectly, the running test of each joint at different speeds (uniform speed every time) was conducted separately, and the oil pressure values in the two chambers of the hydraulic actuator element were measured by the pressure transducer (Model No.U5356-00001U-350BG, range 35 Mpa, 0.1 % accuracy), and then calculated according to Eq. (9) of the equation of motion of the piston of the hydraulic cylinder Get the hydraulic actuator output force.

$$T_i=\begin{cases}[(P_{ai}-P_{bi})-(P'_{bi}-P'_{ai})]V_i & (i=1)\\ 2(S_{ai}P_{ai}-S_{bi}P_{bi}) & (i=2)\\ S_{ai}P_{ai}-S_{bi}P_{bi} & (i=3,5)\\ (P_{ai}-P_{bi})V_i & (i=4,6,7)\,.\end{cases} \tag{9}$$

S_{ai} and S_{bi} are the sensing areas of the rodless chamber piston of the drive hydraulic cylinder of joint i. T_i is the output force/torque of the driving hydraulic actuator of joint i. During the test, it is necessary to adjust the slewing axis of rotary joint 4 to be perpendicular to the horizontal plane; the moving axis of moving joint 3 and joint 5 is adjusted to be parallel to the horizontal plane; the slewing axes of rotary joints 2, 6, and 7 can not be adjusted to be perpendicular to the horizontal plane, and therefore symmetrical operation is required

Table 1. Rubbing model parameters for each joint of 7-DOF HDHM.

Joint i	γ_1	γ_2	γ_3	φ_1	φ_2	φ_3	φ
1	-1800	-694.9	11610	40	11000	5900	-1030
2	-776.4	-339.7	99960	40	11000	290	-583
3	929	862.7	42630	25	250	900	-20
4	264.7	587.2	4515	1	960	2300	-45
5	3609	4095	12660	2.5	700	900	-208
6	209.6	262.6	3426	1	15	2300	-5
7	891.6	918.9	6360	1.5	6	6200	90

for the operation, i.e., running at a uniform speed from $-\theta$ to $+\theta$.

2) Since the axis of rotation of joint 1 and joint 4 is perpendicular to the horizontal plane, the rubbing force is equal to the output torque of the hydraulic actuator at constant speed. The axes of movement of joint 3 and joint 5 are parallel to the horizontal plane, so that the rubbing force is equal to the output force of the hydraulic cylinder at constant speed.

3) Using the rubbing-velocity data of each joint of the 7-DOF HDHM, based on the rubbing model of Eq. (8), the curve fitting tool of MATLAB was used to fit the experimental data to minimize the error between the measured values and the model estimation, and to obtain the optimal parameters of the rubbing model. Finally, the parameter identification results of each joint were obtained.

In order to facilitate the extraction of rubbing features, the parameter identification results of each joint are summarized as shown in Table 1. Finally, the rubbing functions of the seven joints of the 7-DOF HDHM can be obtained $f(\dot{q}_i)$ $(i=1\sim 7)$.

The joint velocity feature in the 7-DOF HDHM raw feature:

$$\boldsymbol{X}_{Vel}=[\dot{q}_1,\dot{q}_2,\dot{q}_3,\dot{q}_4,\dot{q}_5,\dot{q}_6,\dot{q}_7]^{\mathrm{T}}\in\mathbb{R}^{7\times 1}. \tag{10}$$

On the basis of Eq. (10), based on the rubbing function obtained after identification , the JR parameterized features of the MSDM model are calculated and obtained:

$$\boldsymbol{X}_{JR}=[f_1,f_2,f_3,f_4,f_5,f_6,f_7]^{\mathrm{T}}. \tag{11}$$

In Eq. (11), $\boldsymbol{X}_{JR}\in\mathbb{R}^{7\times 1}$ denotes the JR parameterization feature of the MSDM model.

2.2.3 Extraction of joint flexibility (JF) parametric feature

According to the gear joint flexibility model:

$$\begin{cases} i_m T_1 = J_m i_m^2 \ddot{q}_r + \tau_1 \\ \tau_1 = K(q_r - q_1). \end{cases} \tag{12}$$

In Eq. (12), τ_1, T_1 and q_r are the unknown quantities to be solved, K and J_m are constant coefficients. So $i_m^2\ddot{q}_r$ in Eq. (12) is extracted as the JF parameterized feature of the MSDM model:

$$x_{r1}=i_m^2\ddot{q}_r. \tag{13}$$

In Eq. (13), x_{r1} is the joint flexibility parameterized feature of joint 1.

In addition, the flow continuity equation for a valve-controlled hydraulic cylinder is as follows:

$$Q_{ai}=C_{ip}(P_{ai}-P_{bi})+C_{ep}P_{ai}+\frac{S_{ai}(L_{mi}+l_i)}{\beta}\dot{P}_{ai}+S_{ai}\dot{l}_i \tag{14}$$
$$(i=2,3,5)$$

$$Q_{bi}=C_{ip}(P_{bi}-P_{ai})+C_{ep}P_{ai}+\frac{S_{bi}(L_{mi}-l_i)}{\beta}\dot{P}_{bi}+S_{bi}\dot{l}_i \tag{15}$$
$$(i=2,3,5)$$

$$Q_{ai}=C_{im}(P_{ai}-P_{bi})+C_{em}P_{ai}+\frac{V_i(\theta_{mi}+\theta_i)}{\beta}\dot{P}_{ai}+V_i\dot{\theta}_i \tag{16}$$
$$(i=1,4,6,7)$$

$$Q_{bi}=C_{im}(P_{bi}-P_{ai})+C_{em}P_{ai}+\frac{V_i(\theta_{mi}-\theta_i)}{\beta}\dot{P}_{bi}-V_i\dot{\theta}_i \tag{17}$$
$$(i=1,4,6,7).$$

The term containing the modulus of elasticity of the hydraulic fluid, β, represents the effect of hydraulic flexibility on the dynamics of the system, where P_{ai} and P_{bi} are the unknown quantities to be solved and β is a constant factor. Therefore, $S_{ai}(L_{mi}+l_i)$ and $S_{ai}(L_{mi}-l_i)$ in Eqs. (14) and (15), and $V_i(\theta_{mi}+\theta_i)$ and $V_i(\theta_{mi}-\theta_i)$ in Eqs. (16) and (17) are extracted as JR parameterized features of the MSDM model:

$$\boldsymbol{x}_{ri}=\begin{cases} S_{ai}(L_{mi}+l_i) & (i=2,3,5) \\ V_i(\theta_{mi}+\theta_i) & (i=4,6,7) \end{cases} \tag{18}$$

$$\boldsymbol{x}'_{ri}=\begin{cases} S_{bi}(L_{mi}-l_i) & (i=2,3,5) \\ V_i(\theta_{mi}-\theta_i) & (i=4,6,7). \end{cases} \tag{19}$$

$\boldsymbol{x}_{ri}$ and $\boldsymbol{x}'_{ri}$ are the JR parameterized features of joint i.

In summary, the JR parameterization feature of the MSDM model:

$$\boldsymbol{X}_{JF}=[x_{r1},x_{r2},x'_{r2},x_{r3},x'_{r3},x_{r4},x'_{r4},x_{r5},x'_{r5},x_{r6},x'_{r6},x_{r7},x'_{r7}]^{\mathrm{T}}. \tag{20}$$

In Eq. (20), $\boldsymbol{X}_{JF} \in \mathbb{R}^{13\times 1}$ denotes the JR parameterization feature of the MSDM model.

3. Construction of multi-feature semi-parametric dynamics model

The network structure of the MSDM model is shown in Fig. 2, which consists of the following main components: raw data input layer, parametric feature extraction layer, normalization layer, data reorganization layer, RNN layer, random deactivation layer, fully connected layer, inverse normalization layer and output layer.

3.1 Data input and feature extraction layer

The input data of the raw data input layer are the position $\boldsymbol{q} \in \mathbb{R}^{7\times 1}$, velocity $\dot{\boldsymbol{q}} \in \mathbb{R}^{7\times 1}$ and acceleration $\ddot{\boldsymbol{q}} \in \mathbb{R}^{7\times 1}$ of each joint of the 7-DOF HDHM, where the position data are obtained by measurement of the rotary encoder (rotary joint) and the scale (moving joint) mounted on each joint of the 7-DOF HDHM. Velocity data is obtained by first-order differentiation of position data, acceleration data is obtained by second-order differentiation of position data, and non-parametric data is $X_{Ori} \in \mathbb{R}^{21\times 1}$.

The real prediction data of the MSDM model is the joint force/torque of each joint of the 7-DOF HDHM, and the joint force/torque data are obtained by measuring the pressure values obtained from the pressure sensors mounted in the front and rear chambers of the hydraulic actuators of each joint of the 7-DOF HDHM, which are then calculated according to Eq. (9).

The parameterized feature extraction layer is obtained from the RBD, JR, parameter identification and JF models of the 7-DOF HDHM by using the parameterized feature extraction method in the MSDM model building strategy as detailed in Sec. 2.2. Therefore, the 7-DOF HDHM parameterized features include $\boldsymbol{X}_{RBD} \in \mathbb{R}^{7\times 1}$, $\boldsymbol{X}_{JR} \in \mathbb{R}^{7\times 1}$ and $\boldsymbol{X}_{JF} \in \mathbb{R}^{13\times 1}$.

Therefore, the input data for the MSDM model is $\boldsymbol{X}_{Mul} \in \mathbb{R}^{48\times 1}$, as shown in Eq. (21):

$$\begin{cases} \boldsymbol{X}_{Mul} = [\boldsymbol{X}_{Ori}, \boldsymbol{X}_{RBD}, \boldsymbol{X}_{JR}, \boldsymbol{X}_{JF}]^{\mathrm{T}} \\ X_{Ori} = [q_1, \dot{q}_1, \ddot{q}_1, q_2, \dot{q}_2, \ddot{q}_2, \cdots, q_7, \dot{q}_7, \ddot{q}_7]^{\mathrm{T}} \\ \boldsymbol{X}_{RBD} = [\tau_1, T_2, \tau_3, \tau_4, \tau_5, \tau_6, \tau_7]^{\mathrm{T}} \\ \boldsymbol{X}_{JR} = [f_1, f_2, f_3, f_4, f_5, f_6, f_7]^{\mathrm{T}} \\ \boldsymbol{X}_{JF} = [x_{r1}, x_{r2}, x'_{r2}, x_{r3}, x'_{r3}, x_{r4}, x'_{r4}, x_{r5}, x'_{r5}, \\ \qquad x_{r6}, x'_{r6}, x_{r7}, x'_{r7}]^{\mathrm{T}}. \end{cases} \tag{21}$$

The architecture of the MSDM model proposed in this section is shown in Fig. 2. The original input to the MSDM model is still $[\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}]$, but unlike the nonparametric model, $[\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}]$ passes through the feature extraction layer to form a multi-parameterized feature, and then the multi-parameterized feature is fed into the RNN hidden layer.

3.2 Normalization and data restructuring of integrated multi-parameter features

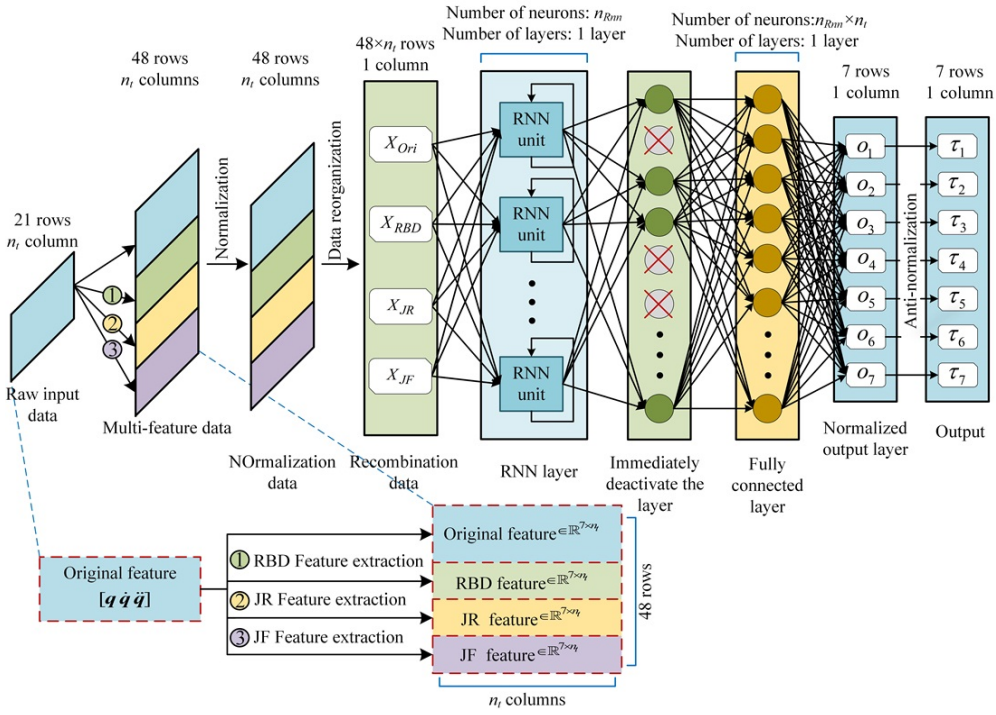


Fig. 2. Network structure of the multi-feature semi-parametric dynamics model of a 7-DOF HDHM.

The integrated multi-parameter features of the MSDM model are normalized, and the normalized integrated multi-parameter features are all in the same order of magnitude, so as to avoid the features with smaller values from being neglected in the training process, and to make the optimization process of the MSDM model smoother and easier to converge to the optimal solution.

In the training of neural networks, data restructuring refers to the process of reorganizing, partitioning, or transforming the original data to meet the input requirements of the model, while enhancing the data's representational capacity and training effectiveness. Specifically, data restructuring typically involves adjustments to the data's dimensions, changes in format, feature combinations, or grouping processes. In the MSDM, the data restructuring layer is responsible for reorganizing the multi-parameter integrated features and training the model in a "many-to-one" fashion. In this process, the restructuring coefficient (n_t) is set to 5, meaning that the data from the first 5 time steps is used as input, and the true joint torque at the 5th time step serves as the output. The sliding step size is set to 1, with a batch size of 256, and the model is trained iteratively. This data restructuring strategy helps to improve data utilization, preserve temporal dynamic features, and reduce the impact of noise.

3.3 RNN layer and fully connected layer

A layer of general recurrent neural network was used for the RNN layer, and the RNN was able to capture the temporal properties in the restructured integrated multiparametric features of the MSDM model, and was able to model the nonlinear time dependencies in the 7-DOF HDHM dynamics. Input to the MSDM model RNN hidden layer of data dimensionality changes, the number of layers of the RNN network and the number of units per layer need to be readjusted according to the dimensionality of the parametric features in the MSDM model. therefore, the effect of different recurrent neural network neuron counts on the computational accuracy of the MSDM model needs to be tested. The feature dimension of the RNN hidden layer of the input MSDM model can be calculated as:

$$n_{Mul} = n_{Ori} + n_{RBD} + n_{JR} + n_{JF} . \tag{22}$$

Then, the number of units in the RNN hidden layer in the MSDM model is:

$$n_{Rnn} = n_{Mul} \times \sigma . \tag{23}$$

In Eq. (23), n_{Rnn} is the number of units in the RNN hidden layer of the MSDM model; n_{Mul} is the dimension of the integrated multi-parameter features; σ is the dimension coefficient, which was trained and tuned, and finally determined to be σ = 2.

The role of the stochastic deactivation layer between the RNN layer and the fully connected layer is to reduce the interdependence between the nodes by randomly zeroing some of the weights or outputs in the RNN layer during the training process of the MSDM model, to achieve the regularization of the RNN layer, so as to reduce the likelihood of overfitting of the MSDM model during the training process, and to improve its prediction accuracy. Each neuron in the fully connected layer is fully connected to all the neurons in its previous layer (a combination of the RNN layer and the stochastic deactivation layer), and it can integrate local information with temporal properties in the RNN layer to improve the performance of the RNN network.

$$n_{fc} = n_{Rnn} \times n_t . \tag{24}$$

In Eq. (24), n_{fc} is the number of neurons in the fully connected layer of the MSDM model; n_{Rnn} is the number of units in the hidden layer of the RNN of the MSDM model, see Eq. (23) for details; n_t is the recombination coefficient, see Sec. 3.2 for details.

3.4 Inverse normalization layer and output layer

In the testing phase of the MSDM model, it is necessary to use the inverse normalization layer to restore the obtained torque values of each joint of the 7-DOF HDHM to the actual order of magnitude, which can then be saved or exported to the manipulator control system for use through the output layer outputs.

3.5 Training and hyperparameter settings of multi-feature semi-parametric dynamics model

Our goal is to predict the force/torque of each joint of the 7-DOF HDHM based on the input $q,\dot{q},\ddot{q}$ of each joint and the extracted parameterized feature data of RBD, JR, and JF, which belongs to the regression prediction study of time series data [28].

The SDM combines the JR and JF features, respectively, resulting in two contrast networks: the SDM+JR and SDM+JF.

In order to evaluate the performance of MSDM in various aspects, the prediction results of MSDM are compared with the four contrasting models (NDM, SDM, SDM+JR, and SDM+JF) to further investigate the effects of the JR and JF parametric features on the inverse dynamics modelling of 7-DOF HDHM. All the dynamics network models presented in this paper are constructed using RNN networks. After many rounds of training and testing, the training and hyperparameter settings of the MSDM model are shown in Table 2.

4. Experimental platform and data collection

7-DOF HDHM experimental platform is constructed to carry

out the operation test under the two operating conditions respectively; 3179500 sets of data are collected and obtained, and the motion data are preprocessed.

4.1 Experimental platform

The experimental platform is shown in Fig. 3 and consists of a mine mill, a 3-tonne liner, a 1.5-tonne liner, a 7-DOF, heavy-duty hydraulic manipulator, a retractable base, and a moveable railcar. The movable railcar moves the manipulator to the mill entrance, and then the telescopic arm transports the manipulator to the inside of the mill to perform the liner change operation. The distribution of the base, seven links, and joints of the 7-DOF HDHM is illustrated in Fig. 4. In Fig. 4, the following components are labeled: L0 (pedestal), L1 (turntable), L2 (main arm), L3 (telescopic arm), L4 (flat swinging wrist), L5 (advancing and retracting wrist), L6 (rolling swinging wrist), L7 (lifting wrist); and the joints are labeled: J1 (large arm rotary joint), J2 (large arm tilting joint), J3 (large arm telescopic joint), J4 (little arm flat swinging joint), J5 (little arm advancing and retracting joint), J6 (little arm rolling swinging joint), and J7 (little arm lifting joint).

Corresponding to the seven motion mechanisms in the 7-DOF HDHM mechanical system, its electro-hydraulic servo drive system mainly consists of the dual-motor drive system for large arm rotary, dual -cylinder synchronous drive system for large arm tilting, large arm telescopic drive system, little arm flat swinging drive system, little arm advancing and retreating drive system, little arm rolling swinging drive system and little arm lifting drive system. Through the electrical signal control electro-hydraulic servo valve spool opening and opening direction, to realize the control of each hydraulic system flow and direction of the robot arm. The hydraulic cylinder parameters of drive joints 2, 3, and 5 are shown in Table 3. The parameters of the hydraulic motors driving joints 1, 4, 6 and 7 are shown in Table 4.

The physical diagram of the control acquisition system and the acquisition operation interface are shown in Fig. 5.

Table 2. Attributes and training settings of RNN-based network of multi-feature semi-parametric dynamics model.

Attributes	Values
Input dimension size	48
Output dimension size	7
Hidden layer	1
Hidden unit	48×σ
Recombination coefficient (n_t)	5
Initial learning rate	0.0018
Drop period of learning rate	60
Drop factor of learning rate	0.5
Epoch	550
Batch Size	256
l_2 -coefficient	0.0002
Dropout rate	0
Optimization solver	Adam

$\sigma = 2$, see Sec. 3.3 for detailed description.

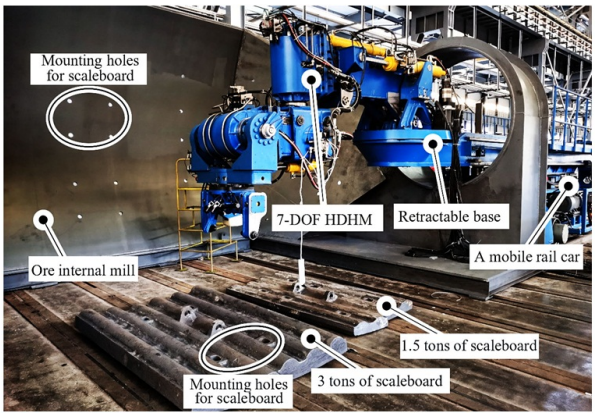


Fig. 3. The experimental platform.

Table 3. Parameters of each hydraulic cylinder.

Parameter	HC_p	HC_t	HC_{fb}
Driving joint	Joint 2	Joint 3	Joint 5
Piston diameter (*mm*)	130	110	100
Piston rod diameter (*mm*)	56	63	56
Travel (*mm*)	790	600	100

HC_p stands for the pitching hydraulic cylinder, HC_t stands for the telescopic hydraulic cylinder, and HC_{fb} stands for the forward and backward hydraulic cylinder.

Table 4. Parameters of each hydraulic motor.

Parameter	HM_r	HM_{hp}	HM_p	HM_l
Driving joint	Joint 1	Joint 4	Joint 6	Joint 7
Displacement (*ml/rev*)	50	2500	1790	2430
Swing angle range (*degree*)	360	±65	±65	±65

HM_r stands for the rotary hydraulic motor, HM_{hp} stands for the horizontal pendulum hydraulic motor, HM_p stands for the pendulum hydraulic motor, HM_l stands for the lift hydraulic motor.

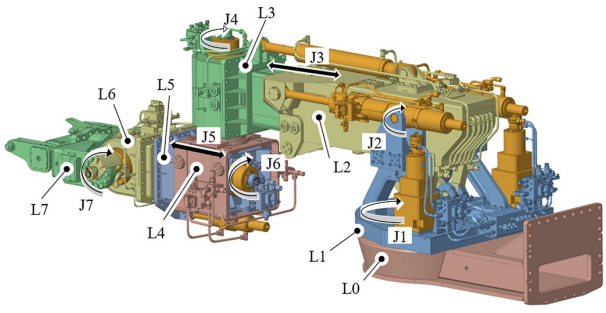


Fig. 4. The structure of the 7-DOF HDHM.

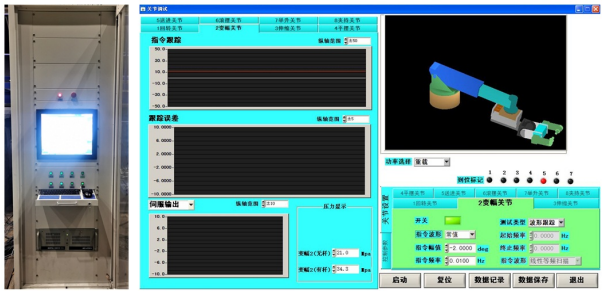


Fig. 5. Physical diagram and operator interface of the control and acquisition system.

4.2 Collection and preprocessing of test data

In order to investigate the dynamics of the 7-DOF HDHM under various operating conditions (especially high-speed and heavy-load conditions), tests were conducted using four operating states (i.e., no-load quick-shift state, gripping state, load quick-shift state, and assembly state). Rotary encoders were used to measure the angular positions of joints 1, 2, 4, 6, and 7, a scale was used to measure the travelling positions of joints 3 and 5, and pressure sensors were used to measure the values of the front and rear chamber pressures of the hydraulic actuators in each joint. The sensor models/specifications are shown in Table 5, and the sampling frequency is 200 Hz. The joint positions of the manipulator and the pressure sensors generate high-frequency noise, so a mean filtering method is used to reduce the noise of the collected test data, and a total of 300000 groups of data are obtained.

7-DOF HDHM Under a load of 3 tons and a maximum operating speed of ≥ 0.2 m/s, 350 movements are completed, and a total of 150000 groups of joint position and hydraulic actuator pressure data are collected, which is called the high-speed and heavy-load data set. The 150000 groups data collected under the load of 1.5 tons and all other conditions are the same, which is called the high-speed and medium-load data set. The motion trajectories were selected based on the following principle: 700 sets of trajectories for load shifting and assembly were selected uniformly in the workspace of the 7-DOF HDHM.

The speed and acceleration limits of each joint are shown in Table 6, the maximum speed and acceleration were used for the tests to make the flexibility and rubbing phenomena of the 7-DOF HDHM more significant. The pressure sensor collects the two-chamber oil pressure values of the hydraulic actuators of each joint of the 7-DOF HDHM, and then the high-frequency noise of the test data is noise reduced using the mean filtering method. The force/torque of each joint was calculated based on the two-chamber oil pressure values of the hydraulic actuators of each joint, the parameters of the hydraulic actuators, and the transmission ratios of some of the joints.

The two datasets were divided into training and test sets according to the ratio of 80 %-20 %, and a 5-fold cross-validation was performed, and then the average value was taken. The exact allocation of the datasets is given in Table 7. All input and output data were normalized using normalization, which helps to speed up the convergence of the model. After the model has been tested, the output is back-normalized to recover its true value.

Table 5. Sensor types/specifications.

Name	Model number	Specification
Pressure sensor	U5335-0100U-350BG	Measuring range 35 Mpa, 0.1 % accuracy; 4-20 mA output
Rotary encoder	RCN2580	Line number 16384, system accuracy ±2.5 um; end at 2.2 s digital output
Grating scale	LC185	Grid distance 20 um, accuracy level ±5 um; End at 2.2 s digital output

Table 6. The speed and acceleration limits of each joint.

Joint i	Speed limit	Acceleration limitation
Joint 1	7 deg/s	3.5 deg/s^2
Joint 2	2.3 deg/s	1.15 deg/s^2
Joint 3	95 mm/s	50 mm/s
Joint 4	16 deg/s	8 deg/s^2
Joint 5	102 mm/s	25 mm/s
Joint 6	17 deg/s	10 deg/s^2
Joint 7	6 deg/s	1.75 deg/s^2

Table 7. Division of training and test sets.

Data set	Training data set	Test data set
High-speed and medium-load	120000 groups	30000 groups
High-speed and heavy-load	120000 groups	30000 groups

5. Experimental results and discussions

Using the obtained data in Sec. 4.4, the five models, namely NDM, SDM, SDM+JR, SDM+JF and MSDM, are fully trained and tested, and their relatively optimal hyper-parameter sets and network structures are obtained and saved respectively. Finally, these optimal network models are validated and the validation results and comparative analyses are given. The network properties and training settings for all models are shown in Table 8. Table 9 lists the RMSE for each joint of the 7-DOF HDHM under the five models.

Analysis of Table 9 shows that the RMSE of each joint (except joints 2 and 4) of the SDM model is slightly reduced compared to the SDM model for the first operating condition of the 7-DOF HDHM, but RMSE of the two models are almost the same. The RMSE of each joint of the SDM+JF model is very close to the SDM model, and RMSE of the two models are very close to each other and is only reduced by 0.46 %. The RMSE of each joint of the SDM+JR and MSDM models were

Table 8. Attributes and training settings of NDM, SDM, SDM+JR, SDM+JF and MSDM models.

Network model	Input dimension size	Output dimension size	Hidden unit
NDM	21	7	$21\times\sigma$
SDM	28	7	$28\times\sigma$
SDM+JR	35	7	$35\times\sigma$
SDM+JF	41	7	$41\times\sigma$
MSDM	48	7	$48\times\sigma$

Detailed attributes and training settings for the five models are set according to the values in Table 2, σ = 2, described in detail in Sec. 3.3.

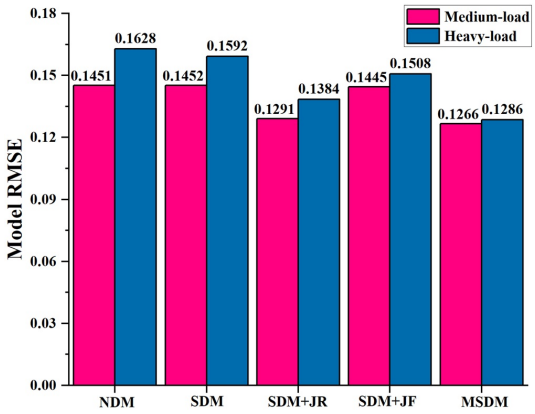


Fig. 6. The RMSE for NDM, SDM, SDM+JR, SDM+JF and MSDM models.

reduced by different magnitudes compared with the SDM model, and RMSE of the two models are approximately 11.04 % and 12.81 % lower, respectively. The RMSE of each joint of the SDM model (except for joints 3 and 7) was slightly reduced compared with that of the SDM model under the second operating condition of the 7-DOF HDHM, and RMSE of the model is approximately 2.21 % lower. The RMSE of each joint of the SDM+JR, SDM+JF, and MSDM models were reduced by different magnitudes compared with the SDM model, and RMSE of the models are approximately 13.07 %, 5.31 %, and 19.20 % lower, respectively.

Fig. 6 shows RMSE of the five models for the 7-DOF HDHM under high-speed and medium-load, and high-speed and heavy-load operating conditions. From Fig. 6, the performance of SDM+JR and SDM+JF models is higher than that of SDM model under the two operating conditions, but lower than that of MSDM model, which verifies the optimal performance of MSDM model.

In terms of computational efficiency, the multi-feature semi-parametric method is compared with the semi-parametric method, and the number of features is improved from 28 to 48 dimensions. Using the trained model to calculate 31810 sets of data, the average time used per round is 65.93 ms (semi-parametric method) and 70.76 ms (multi-feature semi-parametric method), respectively, which is an increase of 7.33 % in computation time. Although the efficiency of the multi-feature semi-parametric method is slightly reduced compared to the semi-parametric method, the computation time for each data set is only 2.22 μs, which still meets the real-time control requirements of the 7-DOF HDHM.

Any 5000 sets of input data are selected and the force/torque data of each joint of the 7-DOF HDHM are predicted using the optimal models of the five models obtained from training and testing, and the force/torque curves of each joint are shown in Figs. 7(a)-(g) are the force/torque curves of the 7 joints for 7-DOF HDHM under the high-speed and medium-load operating conditions. (h)-(n) are the force/torque curves of the 7 joints for 7-DOF HDHM under high-speed and the heavy-load operating conditions.

In Fig. 7, "real" is the actual joint force/torque of each joint obtained by testing as described in Sec. 4.2. "NDM" is the joint force/torque calculated using the NDM model. "SDM" is the joint force/torque calculated using the SDM model. "$MSDM_1$" is the joint force/torque calculated using the SDM+JR model. "$MSDM_2$" is the joint force/torque calculated using the SDM+JF model. "MSDM" is the joint force/torque calculated using the MSDM model.

Analyzing Fig. 7 shows that:

1) The joint force/torque curves of the MSDM model are closest to the actual value curves under the two operating conditions, and the joint force/torque curves of the NDM model are furthest away from the actual value curves.

2) Under the first operating condition, there is no significant difference in the distance between the joint force/torque curves of the NDM model and the SDM model and the actual value. Under the second operating condition, compared with the NDM model, although the joint force/torque curves of the SDM model are closer to the actual values, they both have more obvious gaps. It shows that although the SDM model improves the modeling accuracy of the 7-DOF HDHM under the second operating condition, the improvement is very limited and the effect is not ideal in practical applications.

3) The joint force/torque curves of the MSDM model are higher than those of the SDM+JR, SDM+JF, SDM, and NDM models, in that order, under the two operating conditions. Comparing the curve difference between MSDM model and SDM model, the difference under the second operating condition is larger than the first operating condition. This indicates that the accuracy improvement of MSDM model is more prominent under the second operating condition, and the application of multi-feature semi-parametric method to 7-DOF HDHM dynamics modeling is of practical significance.

The RMSE of different parametric features under the two operating conditions are shown in Table 10. The RMSE of each joint of 7-DOF HDHM and RMSE of the models are given in the table. For comparison, the first two rows show the RMSE of the nonparametric model under the two operating conditions. The analysis shows that the parameterized features affect the prediction results, in descending order, as the JR feature, the JF feature, and the RBD feature.

Table 9. RMSE for each joint of the 7-DOF HDHM under the five models.

Model	Load	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7	Average
NDM	Medium-load	0.2218	0.0879	0.2230	0.0465	0.1504	0.0555	0.1130	**0.1451**
	Heavy-load	0.2045	0.0987	0.2817	0.1038	0.1579	0.0638	0.1218	**0.1628**
SDM	Medium-load	0.2215	0.0995	0.2205	0.0504	0.1485	0.0532	0.1120	**0.1452**
	Heavy-load	0.1996	0.0911	0.2830	0.0749	0.1575	0.0580	0.1242	**0.1592**
SDM+JR	Medium-load	0.1954	0.0893	0.1908	0.0659	0.1456	0.0467	0.0803	**0.1291**
	Heavy-load	0.1937	0.0843	0.2159	0.0731	0.1522	0.0525	0.1076	**0.1384**
SDM+JF	Medium-load	0.2193	0.0923	0.2220	0.0495	0.1486	0.0550	0.1128	**0.1445**
	Heavy-load	0.1975	0.0718	0.2703	0.0561	0.1573	0.0487	0.1076	**0.1508**
MSDM	Medium-load	0.1975	0.0762	0.1921	0.0474	0.1413	0.0449	0.0788	**0.1266**
	Heavy-load	0.1860	0.0603	0.2053	0.0515	0.1491	0.0464	0.0919	**0.1286**

Table 10. Effect of different parameterized features on multi-feature semi-parametric models.

Parameterized feature	Load	RMSE							
		Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Joint 7	Average
NDM	Medium-load	0.2218	0.0879	0.2230	0.0465	0.1504	0.0555	0.1130	**0.1451**
	Heavy-load	0.2045	0.0987	0.2817	0.1038	0.1579	0.0638	0.1218	**0.1628**
NDM+RBD	Medium-load	0.2215	0.0995	0.2205	0.0504	0.1485	0.0532	0.1120	**0.1452**
	Heavy-load	0.1996	0.0911	0.2830	0.0749	0.1575	0.0580	0.1242	**0.1592**
NDM+JR	Medium-load	0.1993	0.0768	0.2002	0.0505	0.1545	0.0542	0.0882	**0.1325**
	Heavy-load	0.1916	0.0722	0.2139	0.0680	0.1537	0.0540	0.1002	**0.1357**
NDM+JF	Medium-load	0.2184	0.0848	0.2236	0.0473	0.1481	0.0537	0.1108	**0.1435**
	Heavy-load	0.1978	0.0662	0.2784	0.0561	0.1558	0.0502	0.1017	**0.1518**

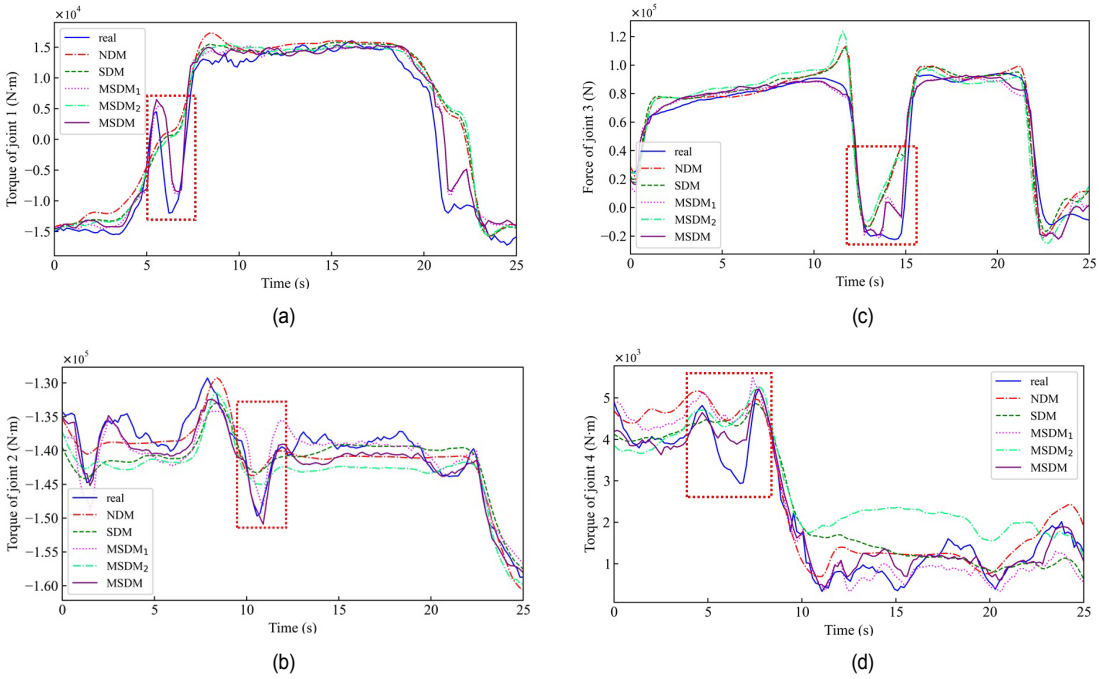


Fig. 7. Comparison of prediction results of NDM, SDM, SDM+JR, SDM+JF and MSDM models for the two operating conditions of 7-DOF HDHM (continue).

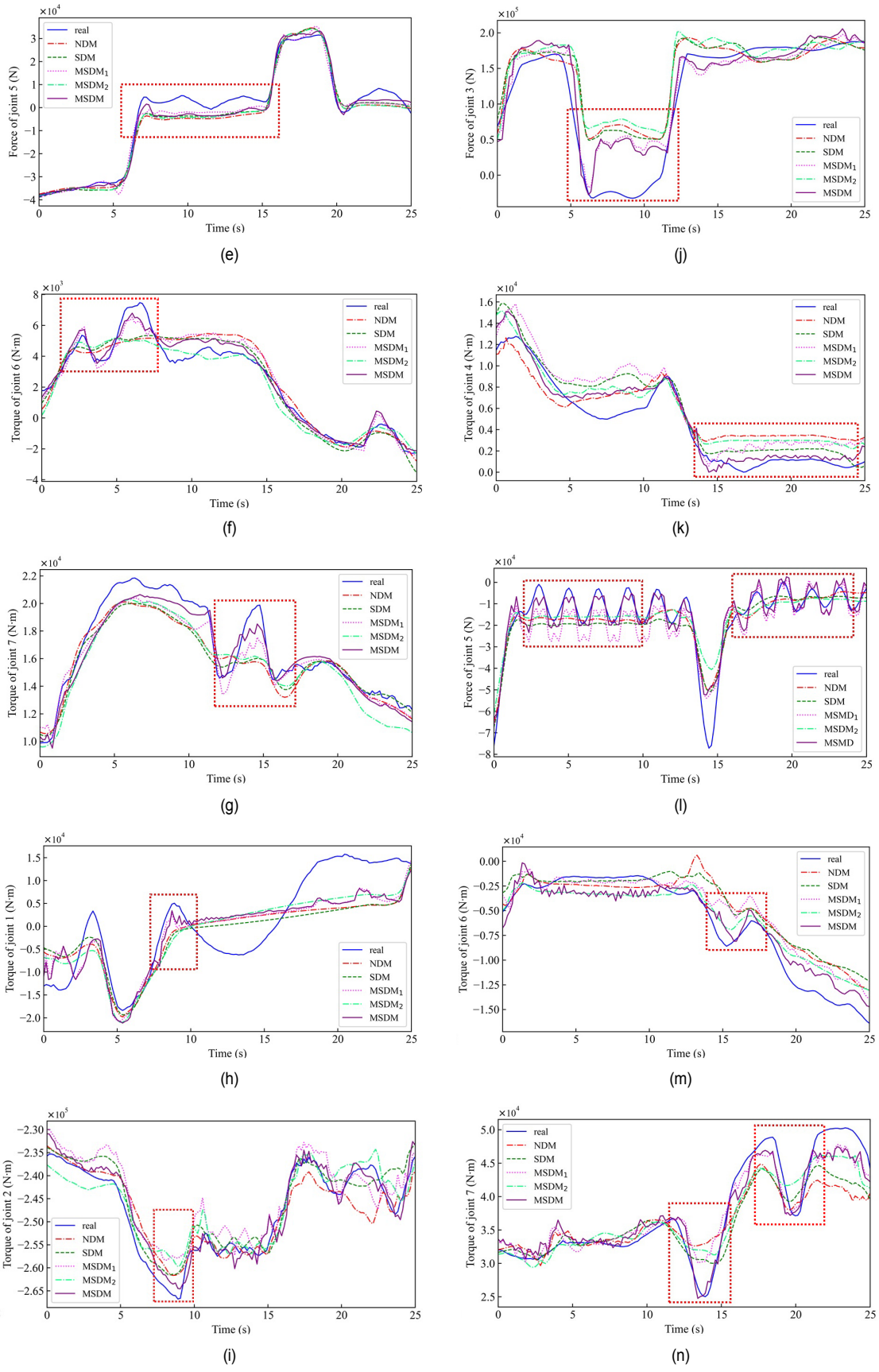


Fig. 7. (Continued).

6. Conclusions

7-DOF heavy-duty hydraulic manipulator (HDHM) is difficult to modeling its inverse dynamics and has low solution accuracy due to the nonlinear factors such as joint rubbing, flexibility deformation, and vibration as well as the coupling characteristics of the control system. In order to solve the above problems, the multi-feature semi-parametric dynamics model (MSDM) is constructed by adopting a hybrid dynamics modeling scheme with deep learning and multi-parameterized features. The model makes full use of the powerful nonlinear fitting ability of deep learning technology, and the mature physical characteristics of rigid body dynamics, joint rubbing, and joint flexibility features. The MSDM model can optimize the non-parametric and parametric parts at the same time, so that the model's prediction accuracy and generalization ability can be significantly improved. The results show that RMSE of the MSDM model is 12.81 % and 19.20 % lower compared with the semi-parametric model under high-speed and medium-load, and high-speed and heavy-load operating conditions, respectively. The MSDM model performs optimally under the two operating conditions, the parameterization features of rubbing and flexibility play a role in the improvement of the 7-DOF HDHM dynamics modeling accuracy. The above research lays the foundation for control and path optimization of the manipulator. In the future, the MSDM model can be further optimized using improved algorithms to obtain better force/torque prediction results. The computation and use of the hydraulic parametric features of the 7-DOF HDHM is also a subject worth exploring. In addition, in this work, the hyperparameters are obtained by manual adjustment, so how to obtain the optimal hyperparameter combinations deserves further study.

Acknowledgments

This work was supported by the National Key Research and Development Program of China (Grant No. 2017YFB1302100, Grant No. 2022YFB3207400).

References

[1] J. Mattila, J. Koivumäki, D. G. Caldwell and C. Semini, A survey on control of hydraulic robotic manipulators with projection to future trends, *IEEE/ASME Transactions on Mechatronics*, 22 (2) (2017) 669-680.

[2] Y. N. Guo, Z. Zhang, Q. Y. Liu, Z. Nie and D. W. Gong, Decoupling-based adaptive sliding-mode synchro-position control for a dual-cylinder driven hydraulic support with different pipelines, *ISA Transactions*, 123 (2022) 357-371.

[3] X. P. Guo, H. S. Wang, L. Wang and H. Liu, Adaptive neural network sliding mode control for serially connected hydraulic cylinders of a heavy-duty hydraulic manipulator, *Journal of Mechanical Science and Technology*, 37 (7) (2023) 3763-3775.

[4] Z. Cui, X. Rong and Y. Li, Design and control method of a hydraulic power unit for a wheel-legged robot, *Journal of Mechanical Science and Technology*, 36 (4) (2022) 2043-2052.

[5] H. Z. Wu and X. Q. Wu, Exploring on application of industrial robot in grinding fastener castings of high speed railway and heavy haul railway, *Railway Engineering*, 59 (7) (2019) 137-141.

[6] R. Q. Ding, J. H. Zhang, B. Xu, M. Cheng and M. Pan, Energy efficiency improvement of heavy-load mobile hydraulic manipulator with electronically tunable operating modes, *Energy Conversion and Management,* 188 (2019) 447-461.

[7] F. L. Zhang and Z. H. Yuan, The study of coupling dynamics modeling and characteristic analysis for flexible robots with nonlinear and frictional joints, *Arabian Journal for Science and Engineering,* 47 (12) (2022) 15347-15363.

[8] Y. M. Song, M. M. Liu, B. B. Lian, Y. Qi, Y. Wang, J. Wu and Q. Li, Industrial serial robot calibration considering geometric and deformation errors, *Robotics and Computer-Integrated Manufacturing,* 76 (2022) 1023-1028.

[9] J. He, H. C. Zheng, F. Gao and H. B. Zhang, Dynamics and control of a 7-DOF hybrid manipulator for capturing a non-cooperative target in space, *Mechanism and Machine Theory,* 140 (2019) 83-103.

[10] L. Li, Z. Y. Lin, Y. Jiang, C. G. Yu and J. Y. Yao, Valve deadzone/backlash compensation for lifting motion control of hydraulic manipulators, *Machines,* 9 (3) (2021) 57.

[11] F. Lixia, T. Wang, Y. Shen, P. J. Wang and M. Wu, Parallel collaborative planning for the coupled system of underground heavy-load robot, *Advances in Mechanical Engineering,* 13 (4) (2021).

[12] X. F. Liu, H. Q. Li, Y. J. Chen and G. P. Cai, Dynamics and control of space robot considering joint friction, *Acta Astronautica,* 111 (2015) 1-18.

[13] B. Li, J. Yan, G. Guo, H. T. Wang and M. J. Zhang, Identification of dynamic parameters and friction coefficients for a heavy-duty hydraulic manipulator, *Proceedings of the 10th World Congress on Intelligent Control and Automation*, Beijing, China (2012) 3102-3106.

[14] J. Smith and M. Mistry, Online simultaneous semi-parametric dynamics model learning, *IEEE Robotics and Automation Letters,* 5 (2) (2020) 2039-2046.

[15] K. H. Low and M. Vidyasagar, A Lagrangian formulation of the dynamic model for flexible manipulator systems, *J. Dyn. Sys. Meas Control,* 110 (2) (1988) 175-181.

[16] J. F. Pan, J. R. Wan, Z. Y. Wang, L. Zheng and L. Y. Jiang, Dynamics modeling of spraying robot using Lagrangian method with co-simulation analysis, *IEEE International Conference on Electrical Engineering and Mechatronics Technology (ICEEMT)*, Qingdao, China (2021).

[17] L. Bascetta, G. Ferretti and B. Scaglioni, Closed form Newton–Euler dynamic model of flexible manipulators, *Robotica,* 35 (5) (2017) 1006-1030.

[18] Y. Zhang, W. H. Ding and H. Deng, Reduced dynamic modeling for heavy-duty hydraulic manipulators with multi-closed-loop mechanisms, *IEEE Access,* 8 (2020) 101708-101720.

[19] K. Yang, W. Y. Yang and C. M. Wang, Inverse dynamic analysis and position error evaluation of the heavy-duty indus-

trial robot with elastic joints: an efficient approach based on Lie group, *Nonlinear Dynamics,* 93 (2018) 487-504.
[20] D. Nguyen-Tuong, M. Seeger and J. Peters, Model learning with local gaussian process regression, *Advanced Robotics,* 23 (15) (2009) 2015-2034.
[21] T. Cederborg, M. Li, A. Baranes and P. Y. Oudeyer, Incremental local online gaussian mixture regression for imitation learning of multiple tasks, *IEEE/RSJ International Conference on Intelligent Robots and Systems*, Taipei, Taiwan (2010).
[22] D. Nguyen-Tuong, J. Peters, M. Seeger and B. Schölkopf, Learning inverse dynamics: a comparison, *European Symposium on Artificial Neural Networks*, Bruges, Belgium (2008).
[23] D. Nguyen-Tuong and J. Peters, Model learning for robot control: a survey, *Cognitive Processing,* 12 (2011) 319-340.
[24] D. Romeres, M. Zorzi, R. Camoriano and A. Chiuso, Online semi-parametric learning for inverse dynamics modeling, *IEEE 55th Conference on Decision and Control (CDC)*, Las Vegas, Nevada, USA (2016).
[25] R. Camoriano, S. Traversaro, L. Rosasco, G. Metta and F. Nori, Incremental semiparametric inverse dynamics learning, *IEEE International Conference on Robotics and Automation (ICRA)*, Stockholm, Sweden (2016).
[26] M. W. Spong, S. Hutchinson and M. Vidyasagar, *Robot Dynamics and Control*, John Wiley and Sons, New York, USA (2008).
[27] C. Makkar, W. E. Dixon, W. G. Sawyer and G. Hu, A new continuously differentiable friction model for control systems design, *Proceedings 2005 IEEE/ASME International Conference on Advanced Intelligent Mechatronics*, Monterey, USA (2005).
[28] Q. Xie, Y. C. Zhang, T. Wang and S. H. Zhu, Dynamic response prediction of hydraulic soft robotic arms based on LSTM neural network, *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*, 237 (7) (2023) 1251-1265.



Nan Liu is a Associate Professor and postgraduate supervisor of Tianjin University of Technology, Tianjin, China. His research interests include robot dynamics, deep learning, etc. He was awarded the First Prize of Tianjin Scientific and Technological Progress, and participated in the project of National Key R&D Programme "Research on Key Technology and Application Verification of Hydraulic Heavy Duty Robotic Arm". He has published more than 20 papers and authorised 6 items.



Xingguo Li is now studying for a master's degree in Tianjin University of Technology, Tianjin, China. His research direction is robot dynamics and deep learning.



Shoujun Wang received the M.Tech. degree in hydraulics and pneumatics from Zhejiang University, in 1989. He is a Professor and doctoral supervisor of Tianjin University of Technology, Tianjin, China. He has published over 100 articles. His current research interests include fluid transmission and control, electromechanical system integration, and unit technology. He led more than 60 research projects sponsored by the Ministry of Transport, the Ministry of Science and Technology, and universities and research institutes, to develop intelligent robots and marine power environment generation technologies. He is a Senior Member of the Chinese Mechanical Engineering Society and the Executive Director of the China Marine Engineering Society.